

**APJ ABDUL KALAM TECHNOLOGICAL UNIVERSITY**

First Semester B.Tech Degree Regular and Supplementary Examination December 2023 (2019 scheme)

**Course Code: EST 100****Course Name: ENGINEERING MECHANICS  
(2019 -Scheme)**

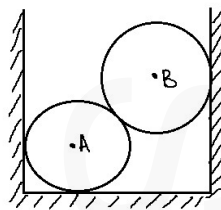
Max. Marks: 100

Duration: 3 Hours

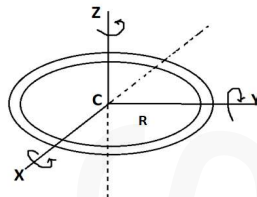
**PART A***Answer all questions, each carries 3 marks*

Marks

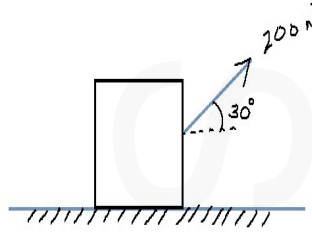
- 1 Explain superposition law and law of transmissibility. (3)
- 2 Two smooth identical spheres A and B are placed in a rectangular channel as shown in figure. Draw the free body diagram of each sphere. (3)



- 3 Distinguish between static friction and dynamic friction. (3)
- 4 A simply supported beam AB of span 5m is carrying point loads 4N, 3N & 5N at 2m, 3m & 4m respectively from support A. Calculate reactions at supports A and B. (3)
- 5 State and prove parallel axis theorem. (3)
- 6 Determine the mass moment of Inertia of a ring about ZZ axis. Density of material –  $\rho$  (3)



- 7 A 25 kg body rests on a horizontal surface for which coefficient of kinetic friction is  $\mu_k = 0.3$ . If the body is subjected to a 200 N force as shown below, determine the acceleration of the body. (3)



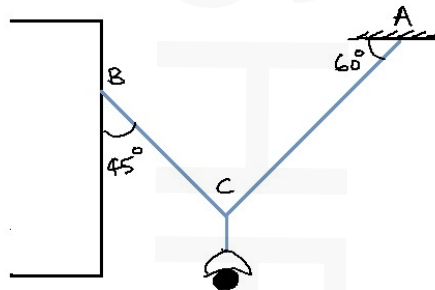
- 8 A body is projected at an angle such that its horizontal displacement is 2 times that of maximum height. Find the angle of projection. (3)
- 9 Explain instantaneous centre of zero velocity. How can you locate it? (3)
- 10 A block of mass 50 kg supported by two springs connected in series hangs from the ceiling. The stiffness of two springs are 4 kN /m and 6 kN /m. Find the equivalent stiffness. (3)

### PART B

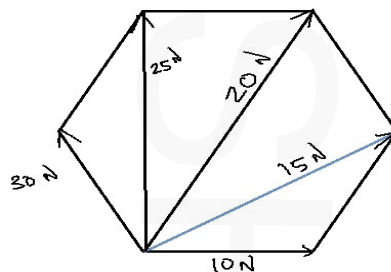
*Answer one full question from each module, each question carries 14 marks.*

#### MODULE 1

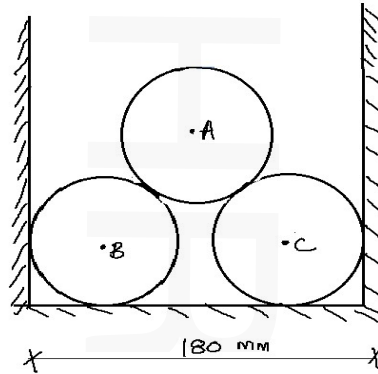
- 11 a An electric light fixture weighing 15 N hangs from a point C, by two strings AC and BC as shown in figure. Determine tension in the strings AC and BC. (7)



- b Forces of 10N, 15N, 20N, 25N and 30N act at an angular point of a regular hexagon towards other angular points as shown in figure. Calculate the magnitude and direction of the resultant force. (7)

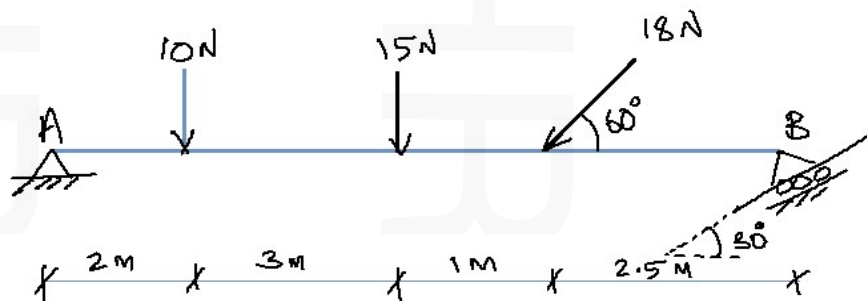


- 12 Three cylinders weighing 100N each and of 80 mm diameter are placed in a channel of 180mm width as shown in figure. Determine the force exerted by (14)
- (i) cylinder A on B at the point of contact. (ii) cylinder B on base (iii) cylinder B on wall.



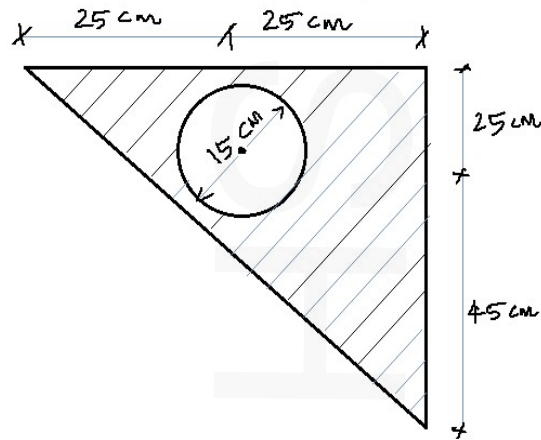
### MODULE 2

- 13 A block weighing 1000 N, overlying a  $15^\circ$  wedge on a horizontal floor and leaning against a vertical wall, is to be raised by applying a horizontal force to the wedge. Assuming the coefficient of friction between all the surface in contact to be 0.3, determine the minimum horizontal force required to raise the block. (14)
- 14 A beam is hinged at A and roller supported at B. It is acted upon by loads as shown below. Find the reactions at A & B. (14)

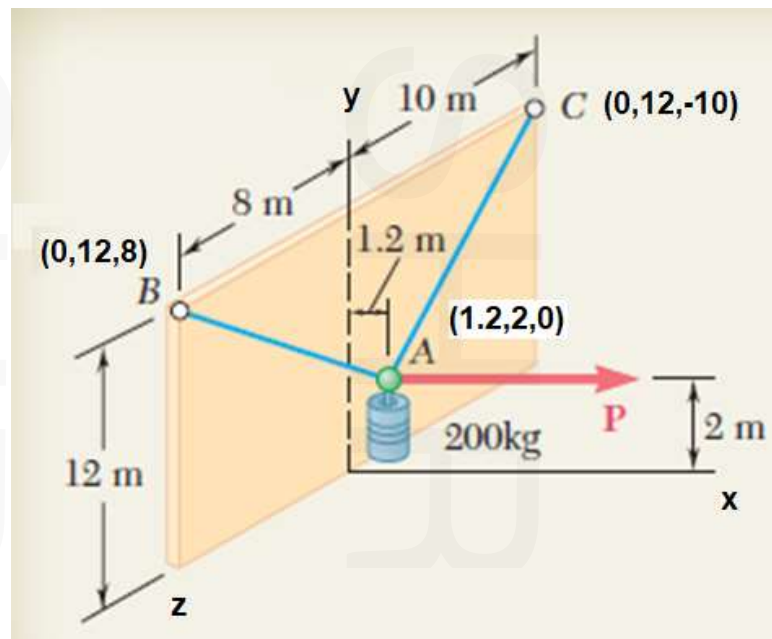


### MODULE 3

- 15 Calculate the moment of Inertia of the shaded area shown in fig, w.r.t. centroidal axes. (14)

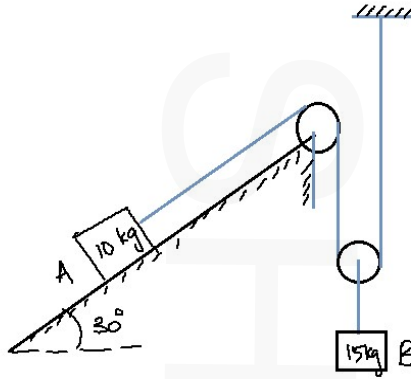


- 16 A 200-kg cylinder is hung by means of two cables AB and AC, which are (14)  
attached to the top of a vertical wall. A horizontal force P perpendicular to the  
wall holds the cylinder in the position shown. Determine the magnitude of P and  
the tension in each cable.



#### MODULE 4

- 17 Determine the resulting motion of the body A of mass 10 kg assuming the (14)  
pulleys to be smooth and weightless as shown in Fig. If the system starts from  
rest, determine the velocity of the body A after 10 seconds. Given coefficient of  
friction as 0.2 and mass of B as 15 kg.



- 18 Find the least initial velocity which a projectile may have so that it may clear a wall 3.6m high and 4.8m distant (from point of projection) and strike the horizontal plane through the foot of wall at a distance 3.6 m beyond the wall. Point of projection is at same level as foot of the wall. (14)

#### MODULE 5

- 19 A flywheel rotates with a constant retardation due to braking. In the first 10 seconds, it made 300 revolutions. At time  $t = 7.5$  sec, its angular velocity was  $40\pi$  rad/sec. Determine (i) value of constant retardation ; (ii) total time taken to come to rest and (iii) total revolutions made till it comes to rest. (14)
- 20 A solid cylindrical pulley of mass 800kg, having 0.8m radius of gyration and 2m diameter is rotated by an electric motor, which exerts a uniform torque of 60kNm. A body of mass 3t is to be lifted by a wire wrapped round the pulley. Find: i) Acceleration of the body & ii) Tension in the rope (14)

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